Ariana Olivares

ariana.olivares@tufts.edu | linkedin.com/in/ariana-olivares-698549185 | sites.google.com/view/arianaolivares

Research Interests: Soft robotics, advanced manufacturing, and the design and control of robotic systems.

EDUCATION

Tufts University, Medford, MA
M.S. in Mechanical Engineering, Controls and Design Track | *GPA: 4.0/4.0* — Expected May 2027
B.S. in Mechanical Engineering, *Magna Cum Laude* | *GPA: 3.61/4.0* — May 2026

RESEARCH EXPERIENCE

M.S. Thesis Research – Nemitz Robotics Group (Sept 2026 – Present)
Tufts University, Department of Mechanical Engineering | Advisor: Dr. Markus P. Nemitz

Injection Molding of Soft Robotic Parts with a Pellet Extruder

- **Objective:** lead an independent study of whether a pellet extrusion head can injection mold simple soft robotic parts, including **ultra-soft elastomers that cannot be reliably 3D printed**.
- **Motivation:** injection molding produces consistent, repeatable parts but requires costly industrial machines; a pellet-extruder approach could give research labs a low-cost way to fabricate soft robotic parts.
- **Approach (in progress):** building on the lab's pellet printing work, adapting a pellet extruder to inject molten elastomer into custom multi-part molds with designed sprues, gates, and vents; tuning melt temperature, screw speed, and hold time to fill molds without voids; benchmarking molded parts against a commercial injection molding machine and against printed and silicone-cast versions.

Undergraduate Research Assistant – Nemitz Robotics Group (Sept 2024 – May 2026)
Tufts University, Department of Mechanical Engineering | Advisor: Dr. Markus P. Nemitz

Pellet Printing for Soft Robotic Devices

- **Objective:** evaluate printing soft robots from ultra-soft elastomer pellets, bridging casting and filament printing.
- **Contributions and products:** fabricated and tested soft robotic parts on the lab's custom pellet printer to assess reliability and durability, analyzing test data in MATLAB and Python; printed actuators from elastomers as soft as Shore 22A withstood **100,000+ inflation cycles** (0–50 kPa) with no failure; designed molds and cast silicone PneuNets as benchmarks; printed a soft robotic fish in one piece; co-authored a 2026 *Advanced Science* paper.

Summer Undergraduate Research Fellow (SURF) – Soft Intelligent Materials Lab (Summers 2024 – 2025)
Stanford University, Department of Mechanical Engineering | Advisor: Dr. Renee Zhao

Freestanding Magneto-Mechanical Metamaterials (Summer 2025)

- **Objective:** build and experimentally validate metamaterials that morph from flat sheets into wave-shaped 3D surfaces, collaborating with Northwestern University, which simulated the multi-wave designs.
- **Contributions and products:** determined how to fabricate and magnetize silicone–NdFeB structures to match simulations, designing fixtures to set each row's magnetization angle (−45° to 45°); built a MATLAB program that **measures 3D deformation from two photos** to compare against simulations; simulated the single-wave design in ABAQUS and MATLAB, matching measured depth **within 1.4%**, with gaps traced to friction and tolerances.

Conformal Magneto-Mechanical Metamaterials (Summer 2024)

- **Objective:** design magnetic metamaterials that conform to and actuate on cylindrical, concave, and convex surfaces.
- **Contributions and products:** designed unit cells with tailored widths and side angles, then molded and magnetized 100+ silicone–NdFeB cells; actuated 12×5 arrays at 50 mT and a 16×3-cell cylinder with an iron-core coil, finding that **raising magnetic particle loading from 25% to 35%** enabled even deployment, validated in ABAQUS.

PUBLICATIONS

Wu, Y., Chen, J.-H., **Olivares, A.**, Kostak, K., Pedicone, S., Kendre, S. V., Nemitz, M. P. *Pellet Printing for Soft Robotic Devices. Advanced Science*, 13(30), e24237 (2026). https://doi.org/10.1002/advs.202524237

POSTER PRESENTATIONS

Olivares, A., Sim, J., Hwang, S., Zhao, R. R. *Freestanding Magneto-Mechanical Metamaterials.* Stanford SURF Symposium, Stanford University, August 2025.
Olivares, A., Wu, Y., Nemitz, M. P. *Pellet Printing for Soft Devices.* Mechanical Engineering Research Symposium, Tufts University, Spring 2025.
Olivares, A., Sim, J., Hwang, S., Zhao, R. R. *Design and Actuation of Conformal Magneto-Mechanical Metamaterials.* Stanford SURF Symposium, Stanford University, August 2024.

TEACHING EXPERIENCE

Teaching Assistant – Senior Capstone Design (Sept 2026 – Present)
Tufts University, Department of Mechanical Engineering | Instructor: Dr. Briana Bouchard

- Advise 40 students through the full design cycle, from requirements and concepts to final design reviews.
- Give technical feedback on CAD models, engineering drawings, analyses, and fabrication plans.

Learning Assistant – Engineering Design (Jan 2025 – May 2025)
Tufts University, Department of Mechanical Engineering | Instructor: Dr. Kristen Wendell

- Mentored 40+ students in SolidWorks, engineering drawings, and prototyping for user-centered design projects.

Learning Assistant – Materials and Manufacturing (Sept 2024 – Dec 2024)
Tufts University, Department of Mechanical Engineering | Instructor: Dr. Luisa Chiesa

- Guided first-year students through CNC, laser cutting, and 3D printing, supervising 15+ final projects.

HONORS AND AWARDS

- **Stanford Engineering SURF Fellow (2024, 2025)** — Selective national summer research program
- **Tufts Engineering Excellence Prize (2025)** — School of Engineering award for research and academics
- **Honos Civicus Society Inductee (2026)** — Tisch College of Civic Life honor for academics and civic leadership
- **SHPE Linde Engineers of Tomorrow Scholar (2025)** — National award for leadership and service
- **SHPE Undergraduate Scholarship (2024)** — National merit-based scholarship
- **YMCA Latino Achiever Scholarship (2022–2025)** — Academic excellence and community engagement

LEADERSHIP AND OUTREACH

Breakthrough San Francisco – Teaching Fellow (June 2026 – Aug 2026)

- Taught 7th grade math, designing lessons on financial literacy, proportional reasoning, and problem-solving.
- Led a daily advisory of students, mentoring them through goal setting, relationship building, and academic support.

Tufts Center for Engineering Education and Outreach – Learning Fellow (May 2023 – Present)

- Teach engineering design, robotics, and programming to 100+ K–8 students through hands-on activities.
- Develop weekly lesson plans and STEM modules tailored to diverse learning levels.

Society of Hispanic Professional Engineers (SHPE), Tufts Chapter – Corporate Liaison (Sept 2022 – Present)

- Organized résumé workshops, mock interviews, and corporate info sessions for engineering students.
- Built partnerships with Google, Microsoft, Vertex, MITRE, GE Aerospace, and Bloomberg.

TECHNICAL SKILLS

Programming: Python, MATLAB, C++, G-code, MicroPython (embedded systems and robotics)
Modeling & Simulation: ABAQUS, SolidWorks, Fusion 360, Onshape | **Languages:** English, Spanish (fluent)
Fabrication: CNC, Injection Molding, Mold Design, Laser/Water-Jet Cutting, 3D Printing (SLA, FDM, pellet)
Materials & Testing: Silicone Casting, Magnetization, UTM and Cyclic Testing, Image-Based Deformation Measurement